# Nicholas Abel

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Software engineer and Unity developer of 10+ years specializing in complex Unity integration challenges and platform specific solutions. Deep experience building Virtual Reality and XR training platforms.

## Skills

Unity (Android, Photon, XR Interaction Toolkit, AR Foundations), C#, .NET (Blazor, Azure Blob Storage, MSAL), JavaScript (d3.js), Meta Quest, glTF, Helm

## Work Experience

### Light & Wonder

**Senior Software Engineer** *Mar 2025 – Feb 2026 | Las Vegas, NV*

Built internal design and 3D model visualization tools for XR, providing an alternative to costly physical mockups and enabling remote feedback and early-stage design reviews.

- Built the XR frontend from scratch in Unity with XR Interaction Toolkit and AR Foundations, targeting Meta Quest 3
- Utilized glTF for runtime model import via Assimp
- Used .NET Azure Blob Storage libraries for remote file management
- Implemented MSAL token flow authentication with a custom Android Keystore wrapper for secure credential storage

**Technologies:** Unity, C#, XR Interaction Toolkit, AR Foundations, Assimp, glTF, Meta Quest 3, Azure Blob Storage, MSAL

### TrainingLab.ai

**Software Engineer (Part-time)** *Sep 2023 – Present*

Building a VR soft-skill training simulation that uses AI to deliver natural, dynamic conversations and assess player responses.

- Created core gameplay systems for goal and task progression tracking
- Integrating AI middleware (Convai) and conversational systems

**Technologies:** Unity, Meta Quest 2, Convai

## AppLovin (Machine Zone)

**Senior Software Engineer** *May 2024 – Oct 2024*

Worked on Veilrunner: Arcane Frontiers, a mobile MMORPG with multi-party tactical combat, challenging boss battles, and a large roster of collectible characters with unique abilities. Position ended on company shutdown.

- Developed Unity Timeline tooling for in-game narrative design and creation
- Added platform-specific lifecycle handling (Android vs. iOS) for player disconnects
- Improved build configuration handling and migrated server-side configs to CI/CD (Helm)

**Technologies:** Unity, Android, iOS, .NET, Orleans, Helm

## Medtronic

**Senior Software Engineer (Contract)** *Dec 2022 – May 2023*

Developed Unity-based VR training modules, including the Mazor Spinal Surgical System. Built internal customer demos and tutorials, and advised on VR training design and methodology.

- Co-architected a VR interaction platform refactor that reduced dependent libraries from 11 to 3
- Integrated platform network code with Photon and built a spawn management system
- Scripted a voice-over generation system using Python and Google Cloud Text-to-Speech

**Technologies:** Unity, Photon, Python, Google Cloud

## Virtualitics

**Software Engineer** *Jun 2021 – Aug 2022 | Pasadena, CA*

Ported a Unity desktop data visualization and analysis tool to a server-side web application.

- One-to-one visual rebuild of the Unity UI in web markup (HTML/CSS)
- Recreated all 2D chart generation in d3.js
- Refactored Unity-dependent libraries to compile outside of the Unity runtime

**Technologies:** C#, .NET, Blazor, JavaScript, d3.js, HTML, CSS

## Osso VR

**Virtual Reality Software Engineer** *Dec 2016 – Jun 2021*

2nd engineer at Osso VR. Turned a prototype VR orthopedic surgical simulator into a product trusted by Medtronic, J&J Medical, and Zimmer Biomet for training and sales enablement, growing from seed funding to $27M (Series B).

- Built out the core interaction system, modularized surgical procedures and specific interactions, and created an OVR abstraction layer

- Created a simulated generic C-arm for procedure accuracy
- Developed VR interaction systems and voice control, tying procedures together in Unity
- Built tools for artists within Unity and updated legacy interaction logic to support multiplayer

**Technologies:** C#, .NET, Unity, Oculus SDK, OpenVR

## Unity

**Unity Developer (Contract)** *Oct 2016 – Dec 2016 | Bellevue, WA*

Worked on various internal VR demonstrations.

- **VR Foundations Toolkit**
  Helped develop the Unity VR Foundations Toolkit (now XR Interaction Toolkit), a collection of Unity scripts for implementing motion controller interactions in VR. Wrote OVR and OpenVR abstractions.
- **EVRTH**
  Implemented globe grab interaction on a simulated planet Earth.

**Technologies:** Unity, C#, OVR, OpenVR

## Tomorrow Today Labs

**Virtual Reality Software Developer** *Aug 2015 – Oct 2016 | Seattle, WA*

Developed Unity-based virtual reality games for the HTC Vive/SteamVR and created physics-based interaction frameworks for motion controller-based VR.

- **NewtonVR**
  Co-created a physics-based interaction system built in Unity3D for SteamVR/HTC Vive, implementing non-kinematic interactions through Unity's physics engine (PhysX)

**Technologies:** Unity, C#, SteamVR, HTC Vive, PhysX

## Microsoft

**Test Associate 1 (Contract)** *May 2010 – Sep 2010 | Redmond, WA*

Ran test cases, filed bug reports, and conducted quality assurance on game titles initially released with the Kinect hardware for the Xbox 360, using a variety of internal tools.

**Technologies:** Xbox 360, Kinect

# Projects

WallPlacer (HoloLens) - Microsoft Holographic Hackathon *2016*

# Publications

Abel, Nicholas (2015). *NewtonVR: Physics-based interaction on the Vive (Part 1)*. VRInflux.com

# Education

## University of Washington – Bothell

**B.S. Computer Science & Software Engineering** *2013 – 2015*

- Bioinformatics Research - Dr. Wooyoung Kim

## Bellevue College

**A.S. Track II (Physics, Atmospheric Sciences, and Engineering)** *2010 - 2012*